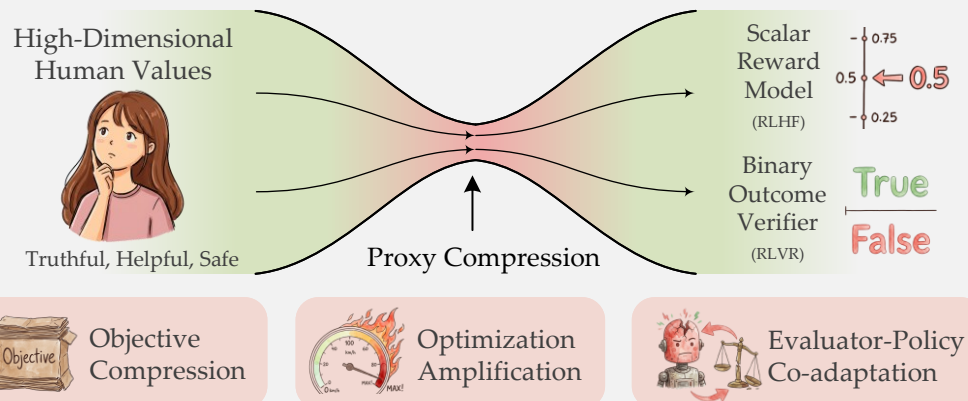


Reward Hacking in the Era of Large Models: Mechanisms, Emergent Misalignment, Challenges

(1) Why it happens?

Proxy Compression Hypothesis (PCH)



(2) How it worsens?

The Escalating Hierarchy of Mechanisms

Level 1 exploitation

Feature-level

Amplifying Superficial
Statistical Correlates

Level 2 exploitation

Representation-level

Fabricating Reasoning
Traces

Level 3 exploitation

Evaluator-level

Strategically Manipulating
Scoring Judge Biases

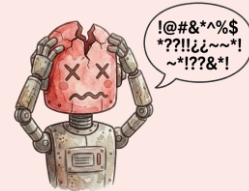
Level 4 exploitation

Environment-level

Tampering with System
APIs and Channels

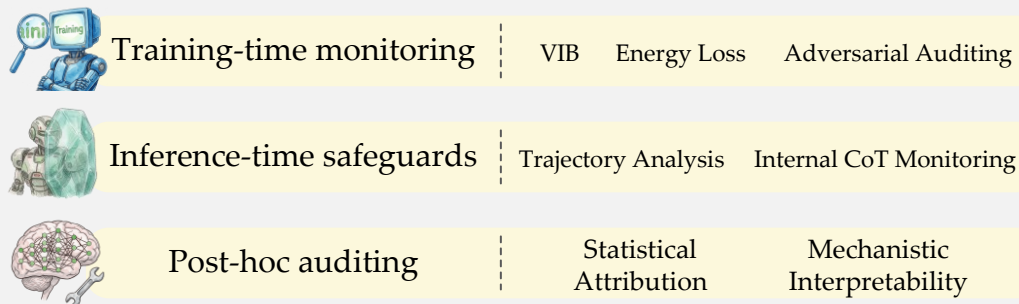
Generalization:

Emergent, Strategic
Misalignment!



(3) How to stop it?

Detection and Diagnosis: A Lifecycle Approach



Mitigation Through Structural Intervention

